Verification Plan Report

Project: AXI-Stream Enterprise Verification

Version: 1.7

Verification Role: Master

1. Test Requirements

Index	Feature/Sub	Name	Description
REQ_HS_01	Handshake (Stability)	Data Stability	Master shall not change TDATA/TUSER while TVALID is HIGH and TREADY is LOW.
REQ_HS_02	Handshake (Timing)	Valid Independence	Master must not wait for TREADY to assert TVALID.
REQ_RST_01	Reset (Suppression)	Reset Zero	TVALID must be driven LOW during active ARESETn.
REQ_RST_02	Reset (Synchronous)	Reset Recovery	First TVALID assertion must be 1+ cycles after reset deassertion.
REQ_PKT_01	Packet (Integrity)	ID Consistency	TID/TDEST must remain constant within a packet (TLAST=LOW).
REQ_PKT_02	Packet (Boundary)	TLAST Signal	TLAST indicates a packet boundary; must be preserved through the system.
REQ_QUAL_01	Qualifiers (Strobes)	TSTRB Validation	TSTRB identifies data bytes; illegal to have TSTRB without TDATA.
REQ_PWR_01	Power (AXI5)	Wakeup Handshake	TWAKEUP must indicate activity and stay HIGH until handshake if concurrent with TVALID.

2. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_HS_01	Valid-Before-Ready: Assert TVALID. Wait 10 cycles for TREADY. Verify bit-stability.	REQ_HS_01, REQ_HS_02
TC_HS_02	Ready-Before-Valid: Assert TREADY. Wait 5 cycles for TVALID. Verify immediate handshake.	REQ_HS_01
TC_HS_03	Master Throttle: Interleave 1-cycle TVALID gaps within a multi-beat packet.	REQ_HS_02
TC_RST_01	Mid-Handshake Reset: Assert ARESETn while TVALID/TREADY are HIGH. Verify TVALID suppression.	REQ_RST_01
TC_RST_02	Reset Exit Timing: Attempt to drive TVALID 0 cycles after ARESETn deassertion. Verify fail.	REQ_RST_02
TC_PKT_01	Standard 16-Beat Packet: Transfer 16 beats with TLAST on the 16th beat.	REQ_PKT_01, REQ_PKT_02

TC_PKT_02	Zero-Data Packet (ZBT): TLAST assertion with all TKEEP/TSTRB bits LOW.	REQ_PKT_02
TC_PKT_03	TID Tampering: Change TID mid-packet without TLAST. Verify assertion.	REQ_PKT_01
TC_PKT_04	TDEST Tampering: Change TDEST mid-packet without TLAST. Verify assertion.	REQ_PKT_01
TC_QUAL_01	Sparse Strobe: Drive packet where only even bytes are valid (TSTRB=0x55/0xAA).	REQ_QUAL_01
TC_QUAL_02	Sideband Stress: Randomized TUSER toggling while TVALID is LOW.	REQ_HS_01
TC_PWR_01	Wakeup Latency: TWAKEUP asserted 3 cycles before first TVALID.	REQ_PWR_01
TC_PWR_02	Wakeup Deassertion Rule: Drop TWAKEUP early while TVALID is waiting. Verify violation.	REQ_PWR_01
TC_STR_01	Randomized Pressure: 10,000 cycles of random TVALID/TREADY/TLAST distributions.	REQ_HS_01, REQ_PKT_01

3. Functional Coverage Groups

Group Name	**Description**
cg_handshake	Exhaustive TVALID x TREADY cross with stall distributions.
cg_packet	Packet lengths 1-4096 beats cross TID change.
cg_strobe	All-ones, all-zeros, and walking-bit TSTRB patterns.

4. Interface Signals

Signal Name	**Direction**	**Width**	**Description**
TDATA	Master	TDATA_WIDTH	Data payload.
TVALID	Master	1	Master valid indicator.
TREADY	Slave	1	Slave ready back-pressure.
TLAST	Master	1	Packet boundary.
TID	Master	TID_WIDTH	Stream identifier.
TDEST	Master	TDEST_WIDTH	Destination identifier.
TUSER	Master	TUSER_WIDTH	Sideband user signals.
TKEEP	Master	TD_W/8	Byte nullify marker.
TSTRB	Master	TD_W/8	Byte qualifier.
TWAKEUP	Master	1	AXI5 Power support.
ACLK	Global	1	Global clock.
ARESETn	Global	1	Global active-low reset.